

14_November_2025_provisional • Unit 1

1H + 2H source-order past paper

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REVIEWED PROVISIONAL | L-to-M

Converted from Linear Higher papers; not an official Pearson paper or mark scheme.

Contents

- 1H Q02 • Rounding, Estimation & Bounds • 2 marks
- 1H Q07 • Rounding, Estimation & Bounds • 2 marks
- 1H Q08 • Algebraic Roots & Indices • 2 marks
- 1H Q10 • Factorising • 7 marks
- 1H Q11 • Right-Angled Triangles - Pythagoras & Trigonometry • 4 marks
- 1H Q13 • Probability Diagrams - Venn & Tree Diagrams • 5 marks
- 1H Q14 • Expanding Brackets • 6 marks
- 1H Q15 • Circles, Arcs & Sectors • 2 marks
- 1H Q19 • Sine, Cosine Rule & Area of Triangles • 3 marks
- 1H Q20 • Coordinate Geometry • 3 marks
- 1H Q21 • Completing the Square • 3 marks
- 1H Q22 • Combined & Conditional Probability • 3 marks
- 1H Q23 • Algebraic Fractions • 3 marks
- 1H Q26 • 3D Pythagoras & Trigonometry • 5 marks
- 2H Q01 • Coordinate Geometry • 2 marks
- 2H Q03 • Fractions • 4 marks
- 2H Q06 • Circles, Arcs & Sectors • 5 marks
- 2H Q07 • Factorising • 3 marks
- 2H Q09 • Probability Toolkit • 3 marks
- 2H Q11 • Right-Angled Triangles - Pythagoras & Trigonometry • 4 marks
- 2H Q13 • Algebraic Roots & Indices • 4 marks
- 2H Q18 • Probability Diagrams - Venn & Tree Diagrams • 3 marks
- 2H Q21 • Histograms • 3 marks
- 2H Q24 • Rounding, Estimation & Bounds • 5 marks

- 2 By rounding each number to one significant figure, work out an estimate for the value of

$$\frac{2.11^2 \times 58.9}{\sqrt{8.859}}$$

Show your working clearly.

(Total for Question 2 is 2 marks)

7 The weight of a bag of apples is 475 g correct to the nearest g

(a) Write down the lower bound of the weight.

..... g
(1)

The height of a box is 120 cm correct to the nearest 10 cm

(b) Write down the upper bound for the height.

..... cm
(1)

(Total for Question 7 is 2 marks)

8 $\frac{8^{-2} \times 8^9}{8^{10}} = 8^n$

Work out the value of n

$$n = \dots\dots\dots$$

(Total for Question 8 is 2 marks)

10 (a) Simplify $(6m)^0$

(1)

(b) Solve $5y + 20 < 7y + 1$

(2)

(c) Factorise fully $15w^2x^5 + 25w^3x^2$

(2)

(d) Simplify fully $\frac{36a^5c^7}{12a^2c^3}$

(2)

(Total for Question 10 is 7 marks)

11 ABC is a right-angled triangle.

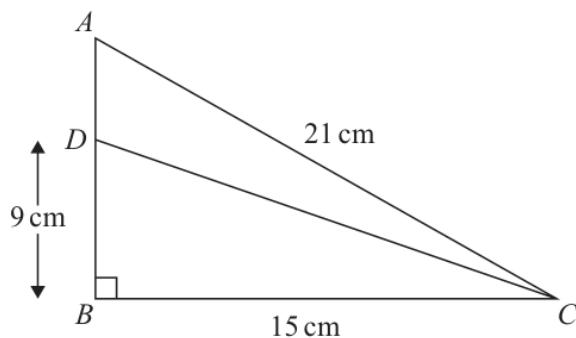


Diagram **NOT**
accurately drawn

$$AC = 21 \text{ cm} \quad BC = 15 \text{ cm} \quad \text{angle } ABC = 90^\circ$$

The point D lies on AB such that $DB = 9 \text{ cm}$

Work out the size of angle ACD

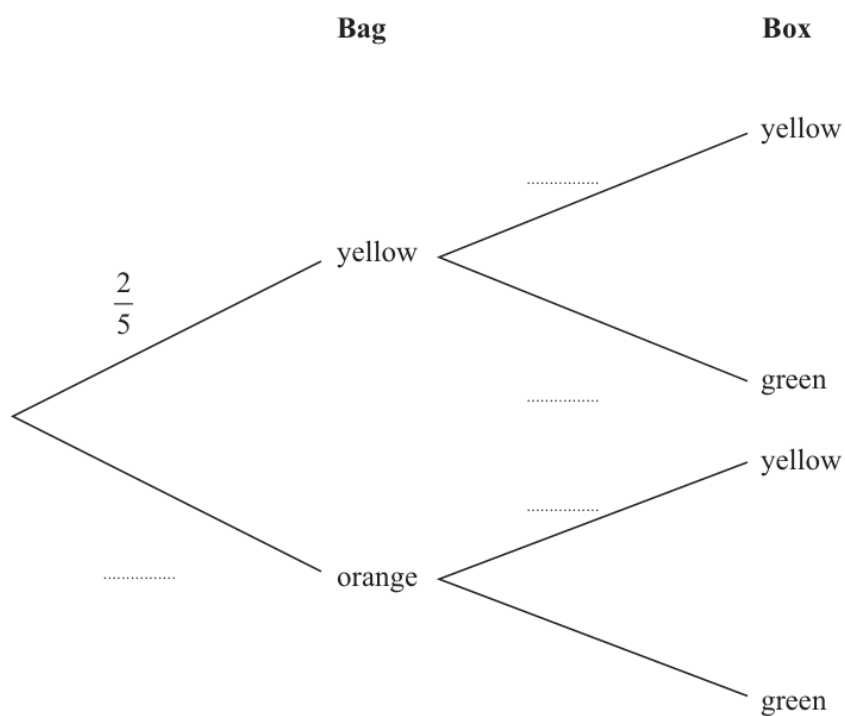
Give your answer correct to one decimal place.

(Total for Question 11 is 4 marks)

- 13 In a bag, there are only 2 yellow beads and 3 orange beads.
In a box, there are only 3 yellow beads and 5 green beads.

Chi takes at random a bead from the bag and a bead from the box.

- (a) Complete the probability tree diagram.



(2)

- (b) Work out the probability that Chi takes two beads of different colours.

(3)

(Total for Question 13 is 5 marks)

14 (a) Expand and simplify $7x(3x + 2)(2x - 5)$

.....
(3)

(b) Solve $\frac{9}{2y} + \frac{5}{7} = 5$

Show clear algebraic working.

$y =$
(3)

(Total for Question 14 is 6 marks)

- 15 AOB is a sector of a circle with centre O and radius 12 cm

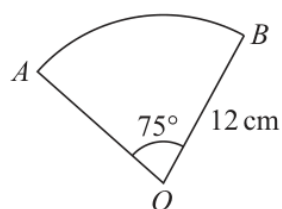


Diagram **NOT**
accurately drawn

Work out the area of the sector.
Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 15 is 2 marks)

19 The diagram shows triangle ABC

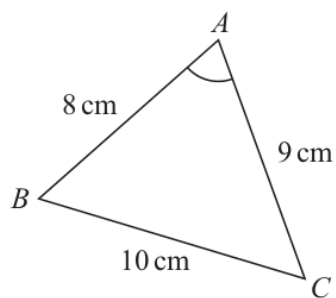


Diagram **NOT**
accurately drawn

$$AB = 8 \text{ cm} \quad BC = 10 \text{ cm} \quad CA = 9 \text{ cm}$$

Work out the size of angle BAC

Give your answer correct to one decimal place.

(Total for Question 19 is 3 marks)

20 The straight line **L** has equation $y = 4x + 7$

The straight line **M** is perpendicular to **L** and passes through the point with coordinates (8, 1)

Find an equation for **M**

Give your answer in the form $y = mx + c$

(Total for Question 20 is 3 marks)

21 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a , b and c are integers.

(Total for Question 21 is 3 marks)

22 There are 15 buttons in a box.

- 3 of the buttons are red
- 2 of the buttons are pink
- 10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

(Total for Question 22 is 3 marks)

23 $a = \frac{2x+5}{1-x}$ $x = \frac{5-2y}{3y}$

Write a in the form $\frac{m+ny}{p(y-1)}$ where m , n and p are integers.

Show your working clearly.

$a = \dots\dots\dots$

(Total for Question 23 is 3 marks)

26 Here is a prism $ABCDEFGH$ with a horizontal, rectangular base $ABGF$

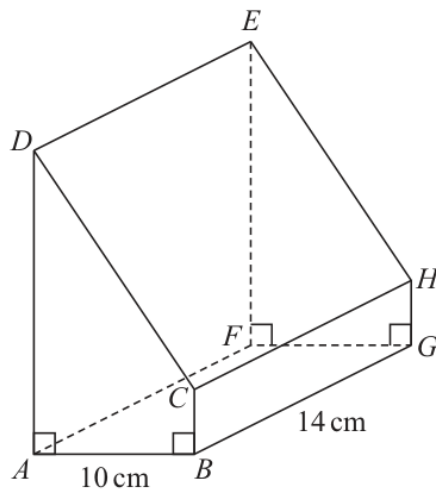


Diagram **NOT**
accurately drawn

$$AB = 10 \text{ cm} \quad BG = 14 \text{ cm}$$

$$\text{angle } DAB = \text{angle } ABC = \text{angle } EFG = \text{angle } FGH = 90^\circ$$

$$BC = \frac{1}{5} AD$$

The angle of elevation of E from B is 50°

Calculate the volume of the prism.

Give your answer correct to 3 significant figures.

Show your working clearly.



..... cm³

(Total for Question 26 is 5 marks)

- 1 The point P has coordinates $(3, 4)$
The point Q has coordinates $(9, 16)$

M is the midpoint of the line PQ

Find the coordinates of M

(..... ,)

(Total for Question 1 is 2 marks)

3 A box contains only

9 red bricks
43 blue bricks
and some yellow bricks

$\frac{7}{20}$ of the bricks are yellow bricks.

Each brick weighs 35 grams.

Work out the total weight of the yellow bricks.

..... grams

(Total for Question 3 is 4 marks)

- 6 A large circle with centre O contains 3 identical small circles. Each circle touches two other circles.

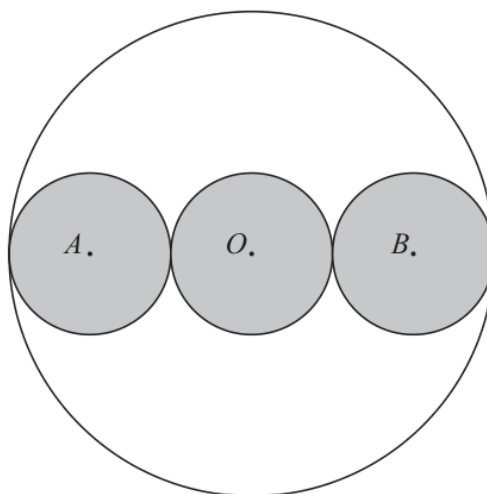


Diagram **NOT**
accurately drawn

A , O and B are the centres of the small circles.
 AOB is a straight line.

The circumference of the large circle is 160 cm

Work out the area of one small circle.
Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 6 is 5 marks)

7 (i) Factorise $x^2 + 2x - 48$

.....
(2)

(ii) Hence, solve $x^2 + 2x - 48 = 0$

.....
(1)

(Total for Question 7 is 3 marks)

- 9 Mark has three bags, bag A, bag B and bag C

Bag A contains 120 counters.

The probability of taking at random a red counter from bag A is 0.45

Bag B contains 80 counters.

The probability of taking at random a red counter from bag B is 0.3

Bag C is empty.

Mark puts all the counters from bag A and bag B into bag C

Mark takes at random a counter from bag C

Work out the probability that he takes a red counter.

(Total for Question 9 is 3 marks)

- 11 The diagram shows a right-angled triangle ABC

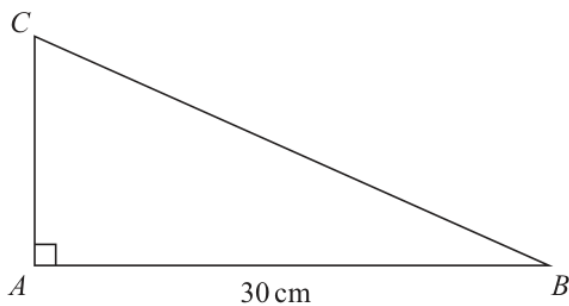


Diagram **NOT**
accurately drawn

Area of triangle $ABC = 240 \text{ cm}^2$

Work out the perimeter of triangle ABC

..... cm

(Total for Question 11 is 4 marks)

13 (a) Simplify fully $\left(\frac{125x^6}{y^{15}}\right)^{\frac{2}{3}}$

.....
(2)

Given that $4^n = \frac{4^m}{64^p}$

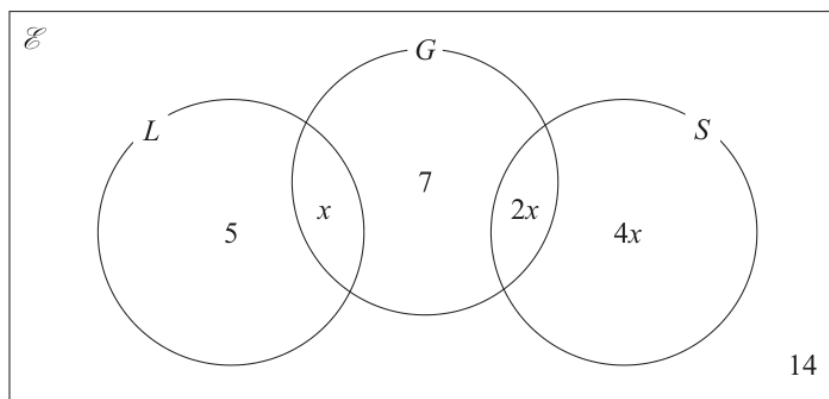
(b) Express n in terms of m and p

$n =$
(2)

(Total for Question 13 is 4 marks)

- 18 Bianca asked 40 students which of the languages Latin (L), Greek (G) and Sanskrit (S) they study.

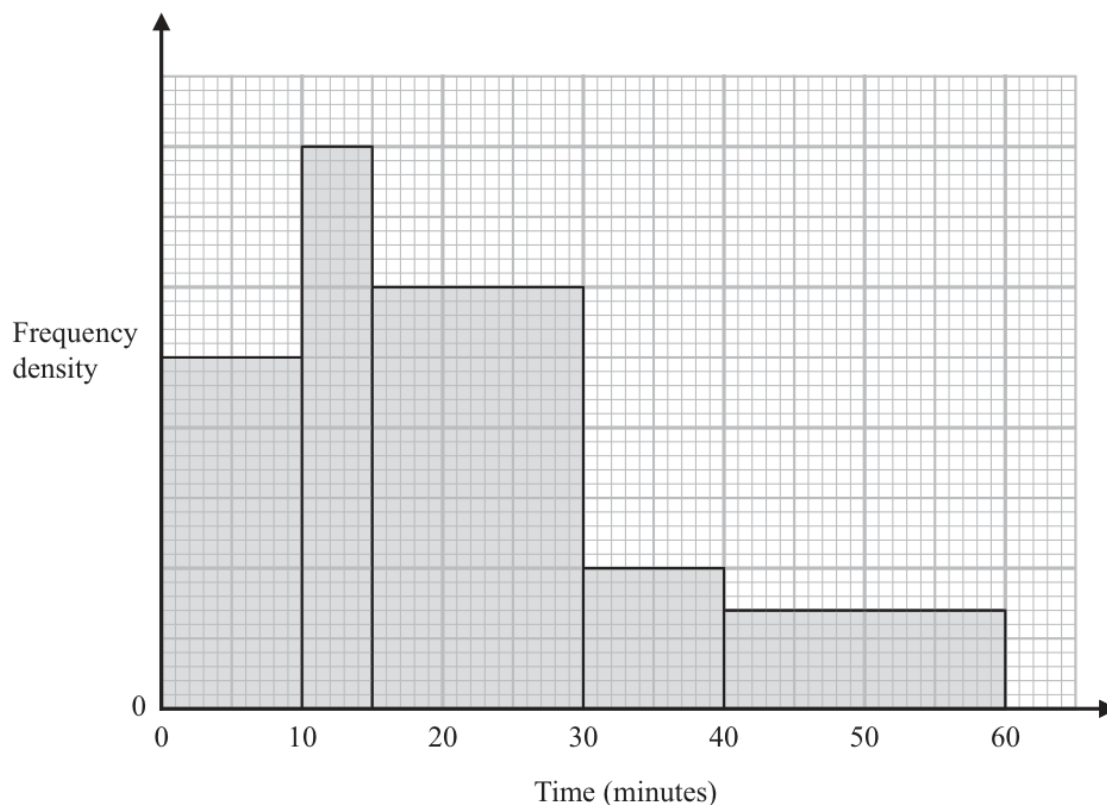
The Venn diagram gives some information about her results.
It shows the number of students in each subset.



One of these students is selected at random.
Work out the probability that this student studies Latin or Sanskrit.

(Total for Question 18 is 3 marks)

- 21 The histogram shows information about the times, in minutes, that some trains arrived late at a station one day.



20 of these trains arrived between 10 minutes late and 15 minutes late.

No trains arrived more than 60 minutes late.

Work out an estimate for the number of these trains that arrived at least 25 minutes late.

(Total for Question 21 is 3 marks)

24 $D = \frac{n}{p - q}$

$n = 10.3$ correct to 1 decimal place

$p = 7.24$ correct to 2 decimal places

$q = 4.39$ correct to 2 decimal places

By considering bounds, work out the value of D to a suitable degree of accuracy.
Show your working clearly.

(Total for Question 24 is 5 marks)